

Joseph Fourier (1768-1830)

Fourier was obsessed with the physics of heat and developed the Fourier transform theory to model heat-flow problems.

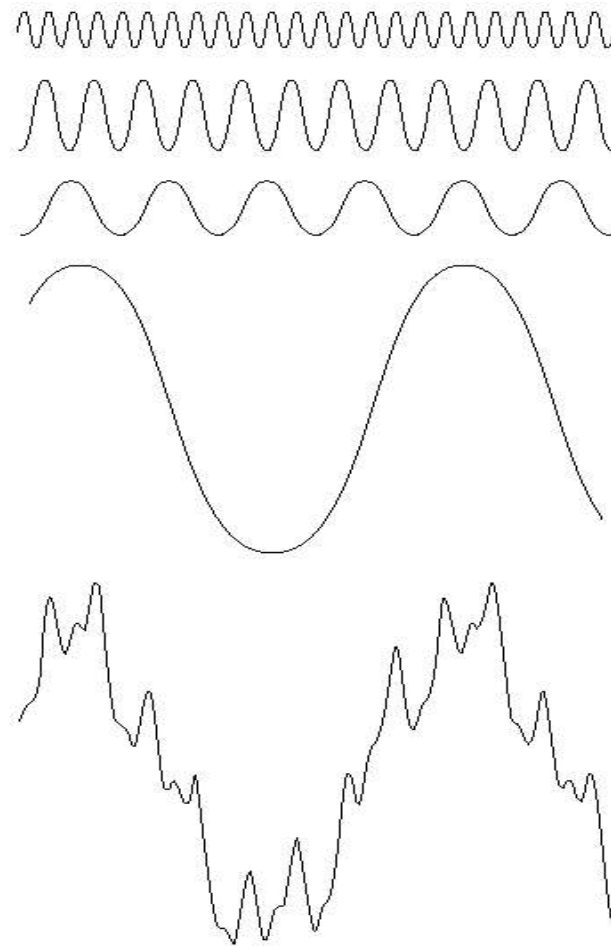


Joseph Fourier, 21 March 1768–16 May 1830. (By permission of the Bibliothèque Municipale de Grenoble.)

Fourier Transform

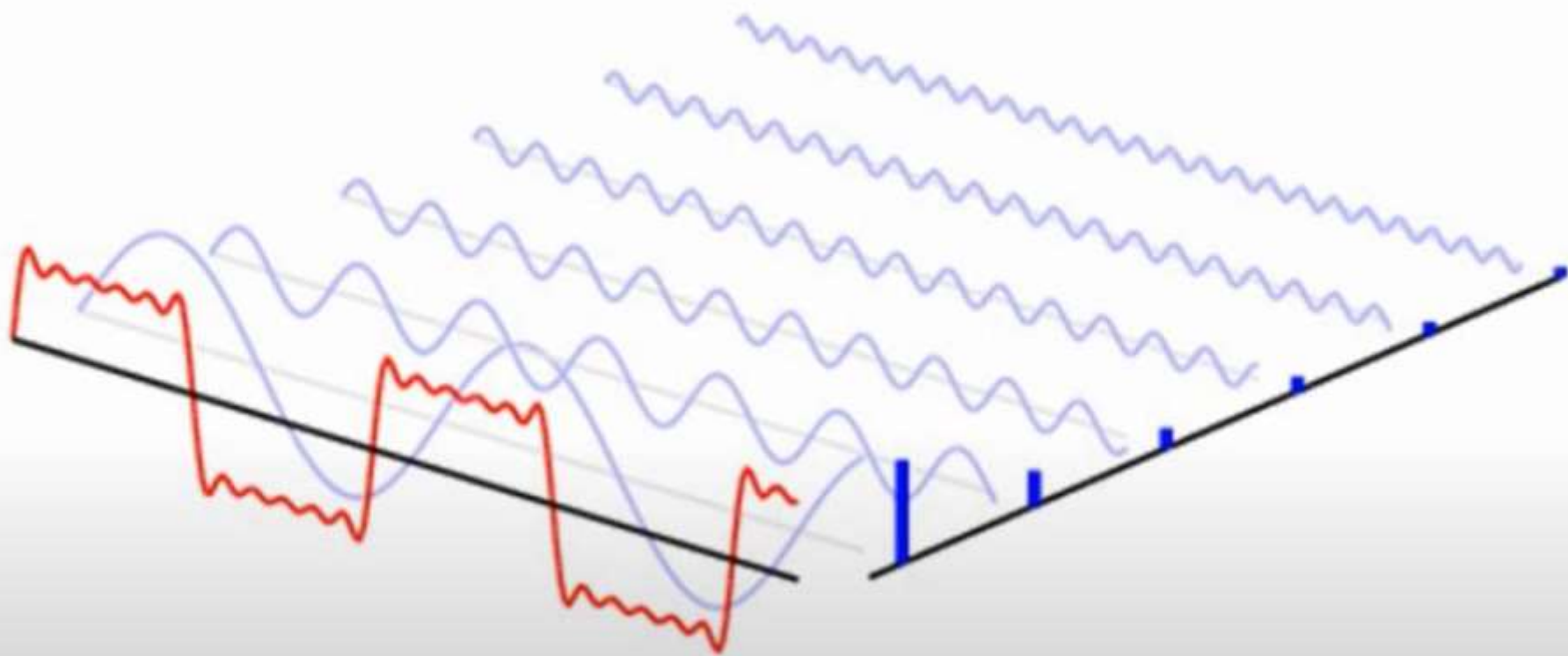
Basic ideas:

- A periodic function can be represented by the **sum** of sines/cosines functions of different frequencies, multiplied by a different coefficient.
- Non-periodic functions can also be represented as the **integral** of sines/cosines multiplied by weighing function.



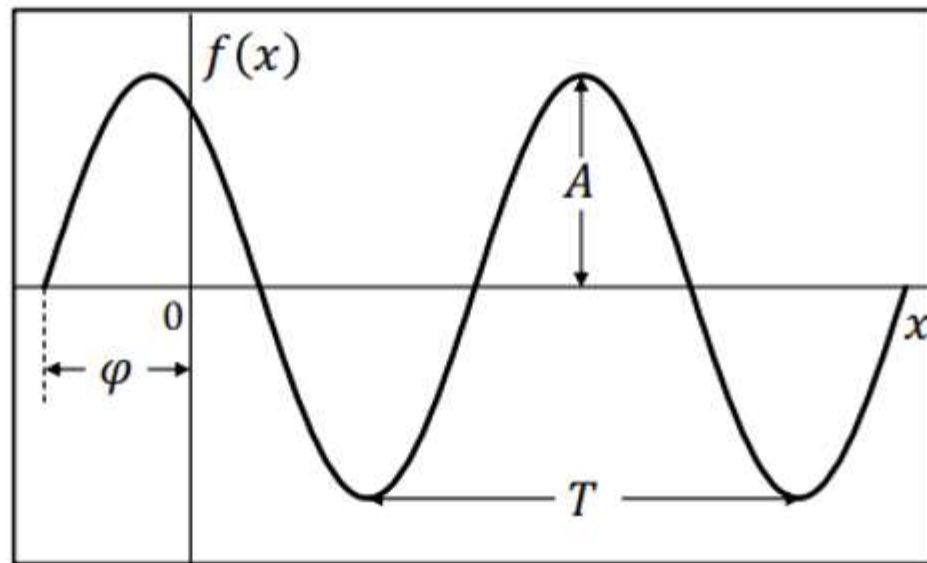
Fourier Transform





Sinusoid

$$f(x) = A \sin(2\pi u x + \varphi)$$



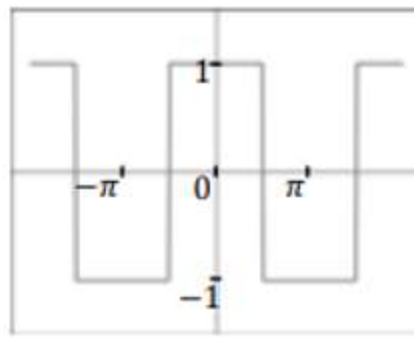
A : Amplitude

T : Period

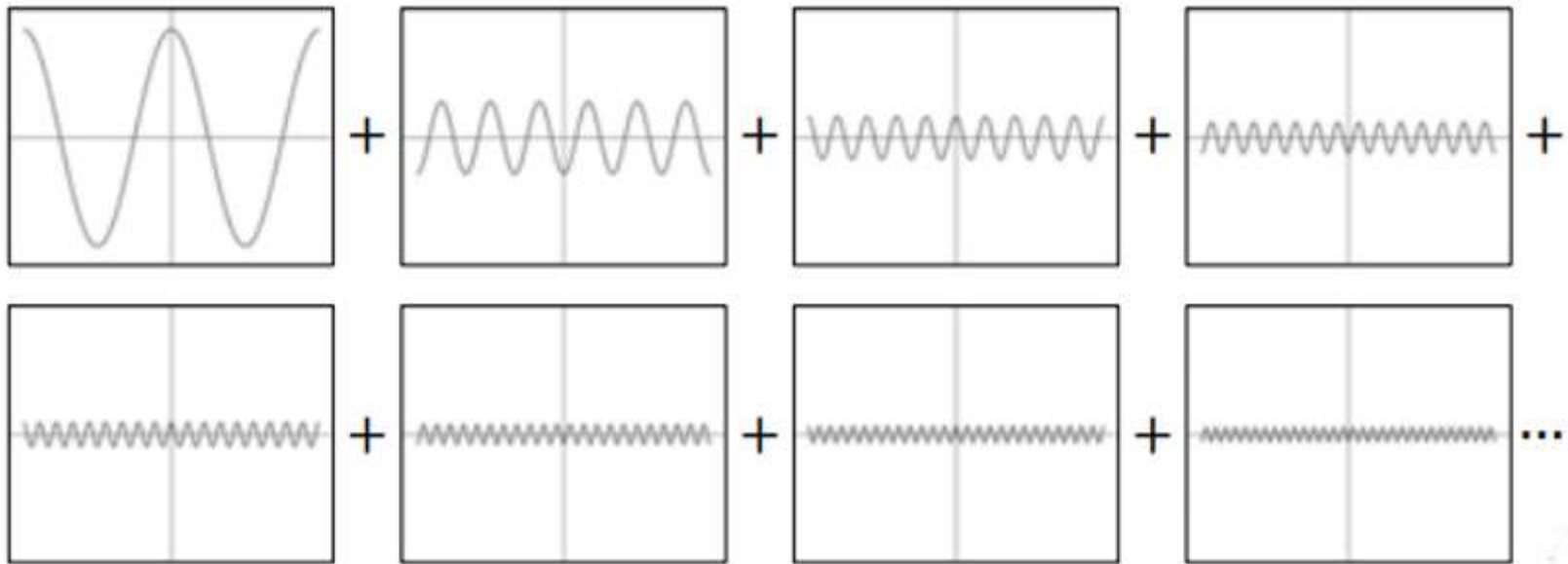
φ : Phase

u : Frequency ($1/T$)

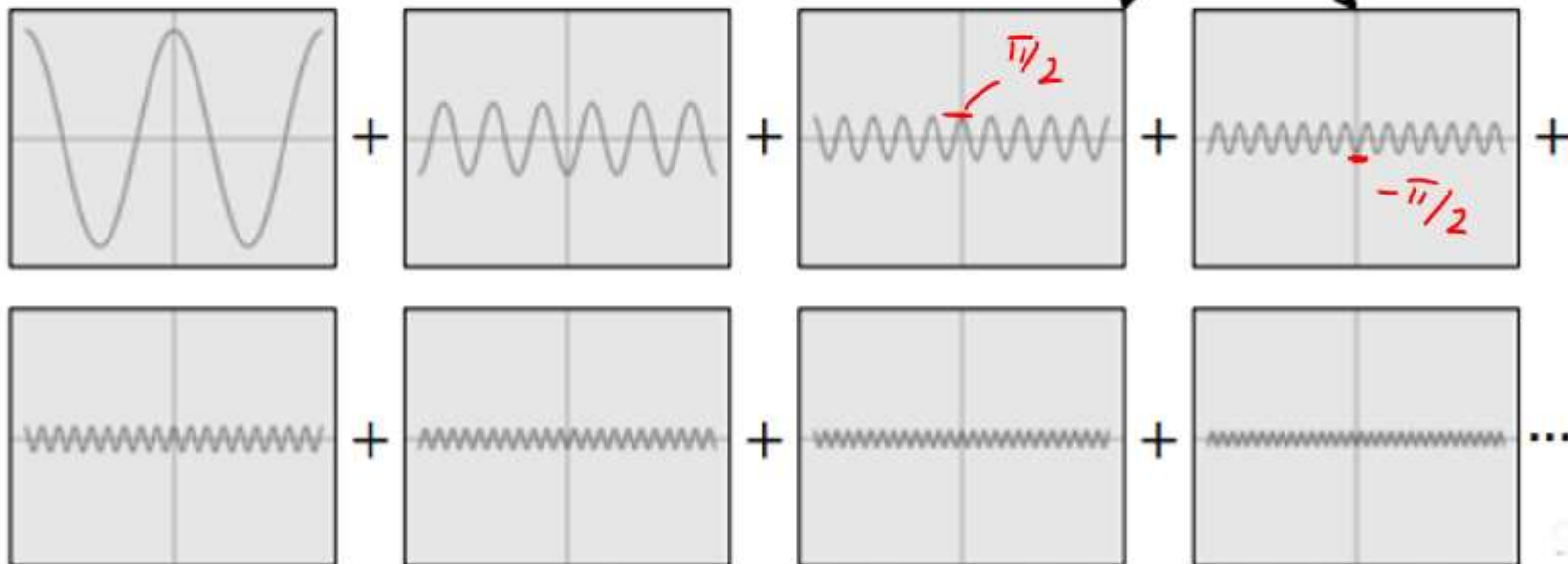
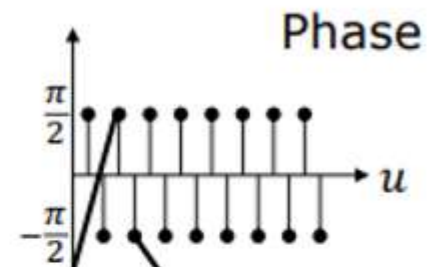
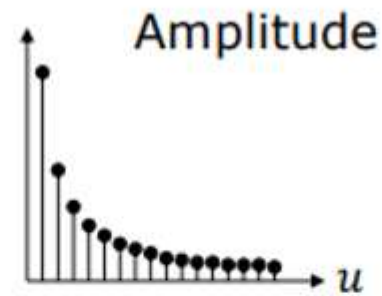
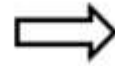
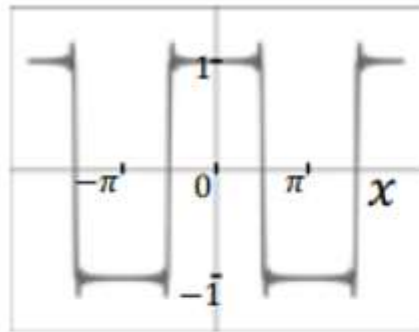
Fourier Series

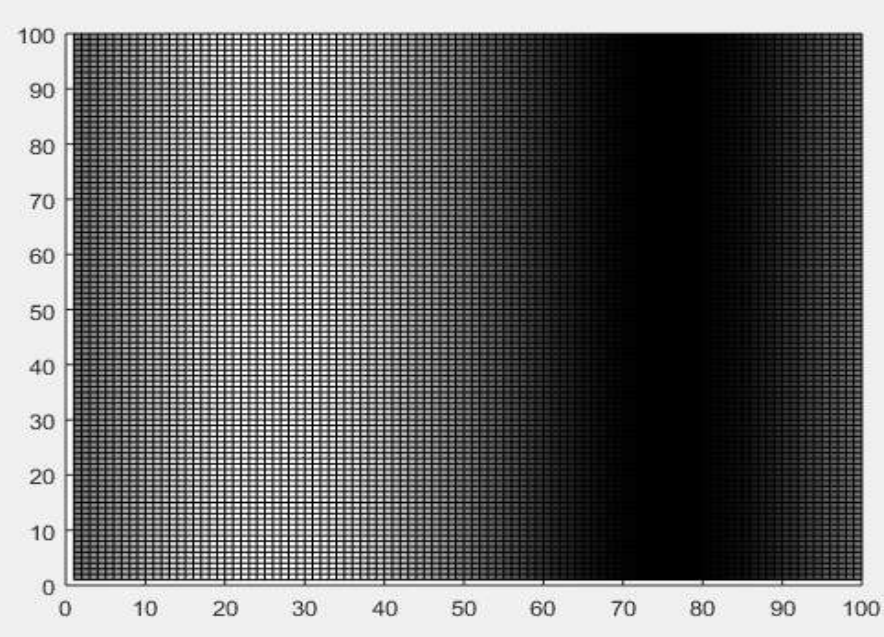
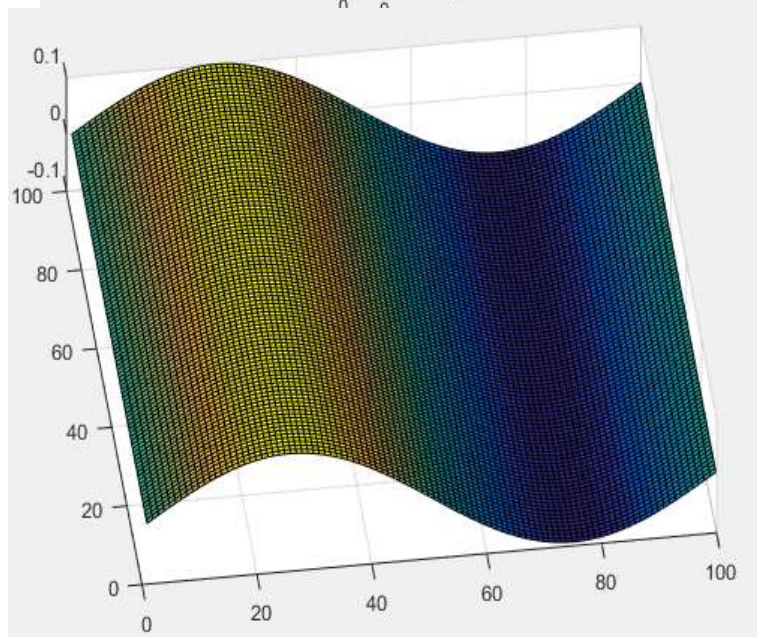
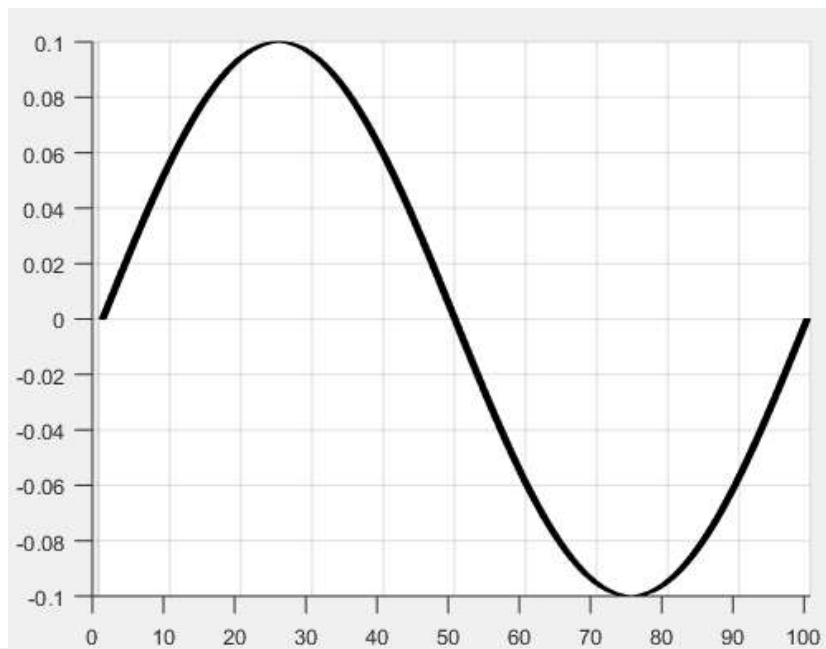
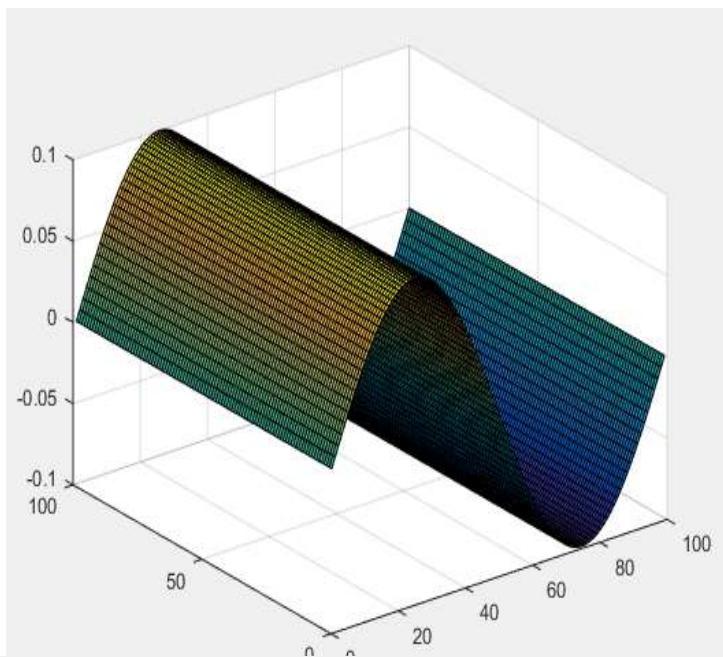


— Square Wave
(Period 2π)



Frequency Representation of Signal

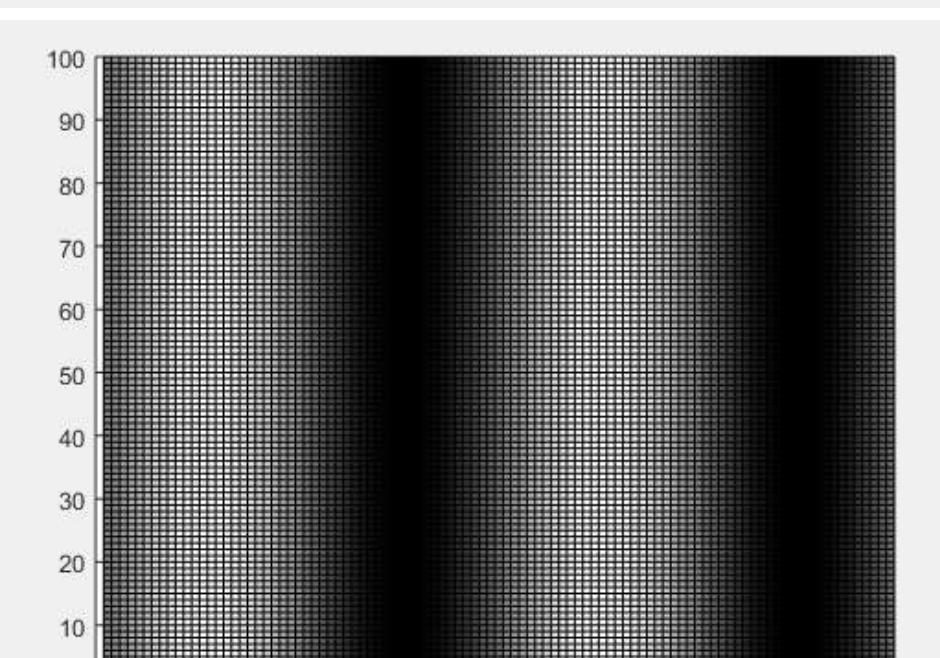
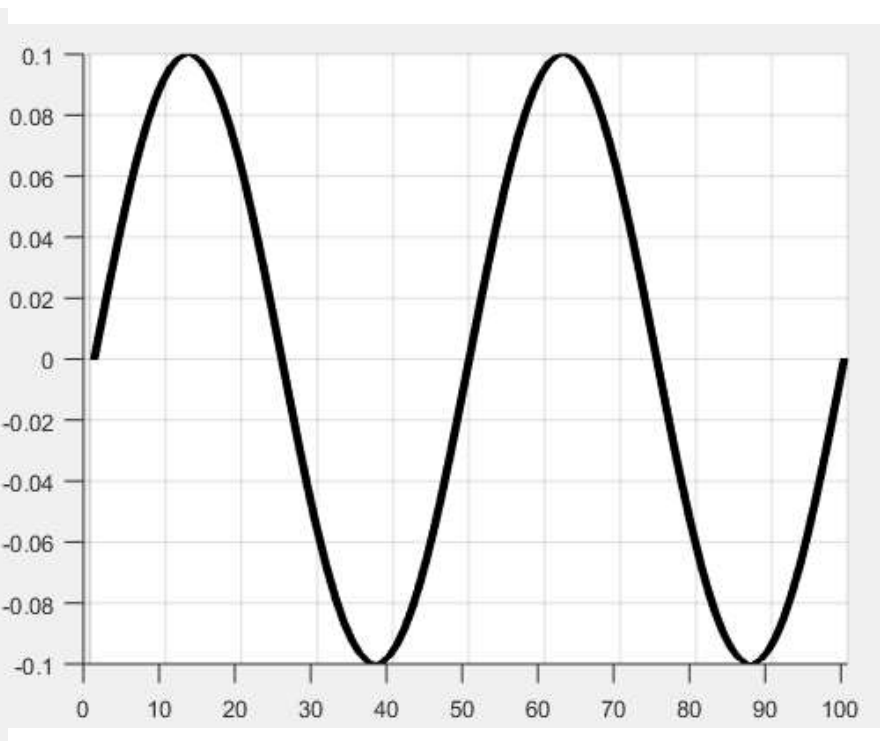
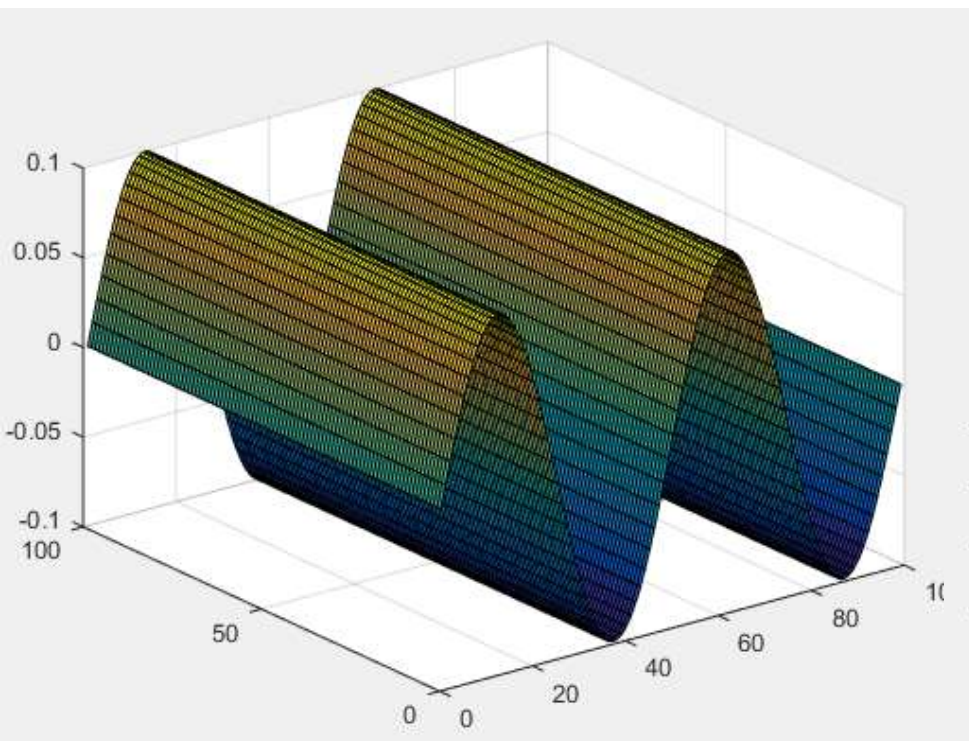


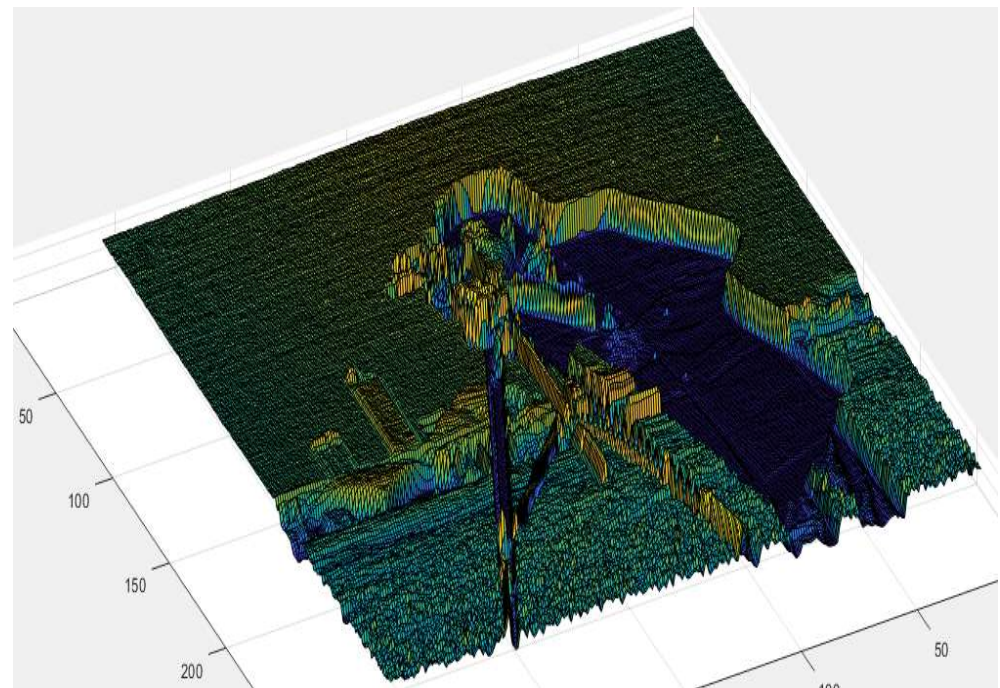
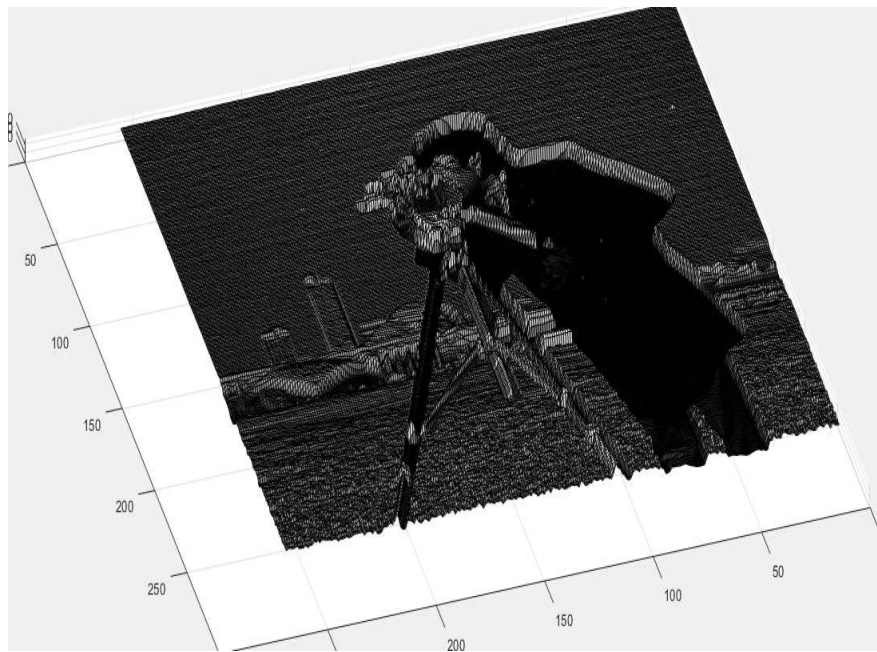



```
i=linspace(0,2*pi);  
p=zeros(100,100);  
j=i;
```

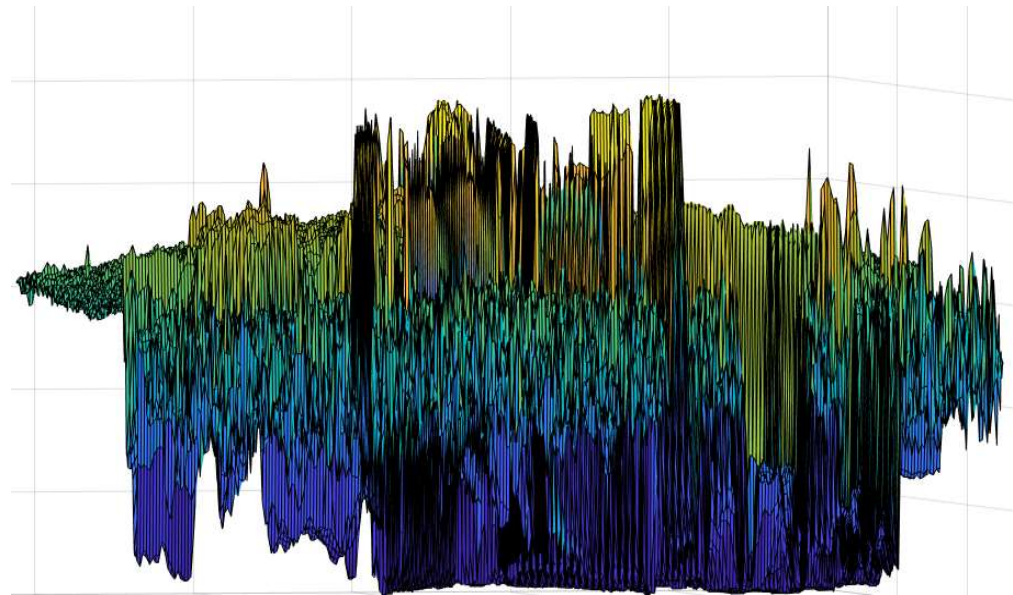
```
- for m=1:1:100  
-   for n=1:1:100  
        p(m,n)=0.1 *sin(i(1,n));  
        % p(m,n)=p(m,n)+ 0.1 *cos(k*i(1,n));  
    end  
end
```

```
figure,surf(p);  
colormap(gray);
```





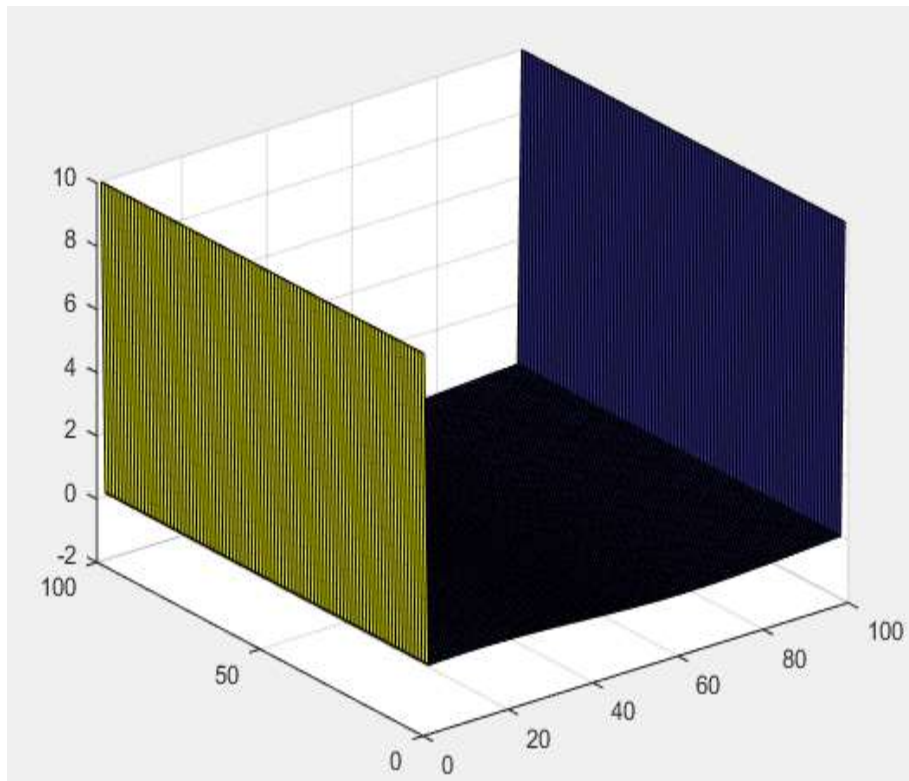
```
a=imread('cameraman.tif');  
a=double(a);  
surf(1:256,1:256,a);  
,
```



```
i=linspace(0,2*pi);  
p=zeros(100,100);  
j=i;
```

```
- for m=1:1:100  
-   for n=1:1:100  
  
        p(m,n)=0.1 *sin(2*i(1,n));  
        %   p(m,n)=p(m,n)+ 0.1 *cos(k*i(1,n));  
    end  
end
```

```
figure,surf(p);  
colormap(gray);
```

```

i=linspace(0,2*pi);
p=zeros(100,100);
j=i;
for k=1:1:100
    for m=1:1:100
        for n=1:1:100

            p(m,n)=p(m,n)+0.1 *sin(k*i(1,n));
            p(m,n)=p(m,n)+ 0.1 *cos(k*i(1,n));

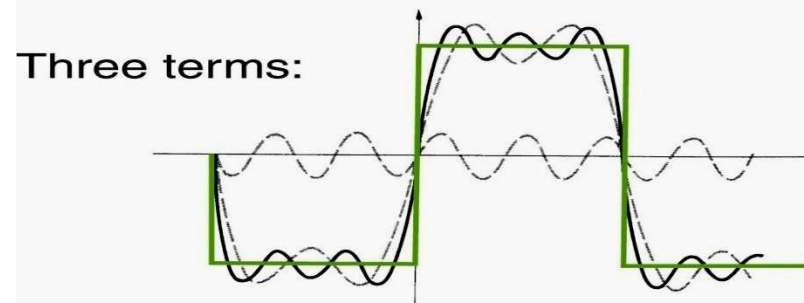
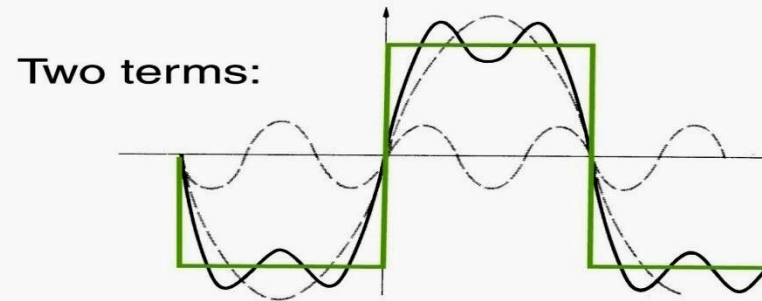
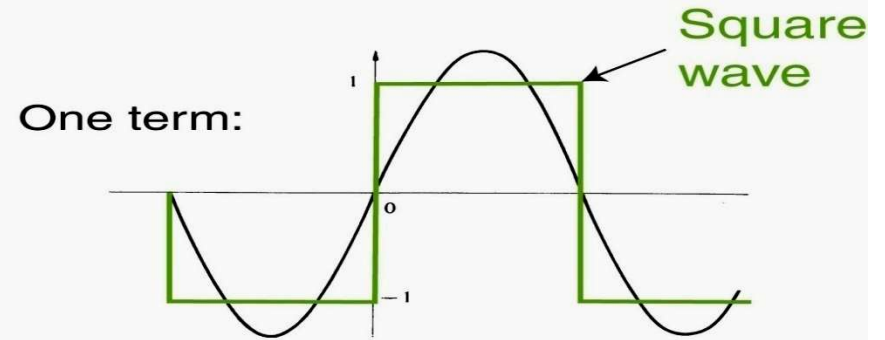
        end
    end
end

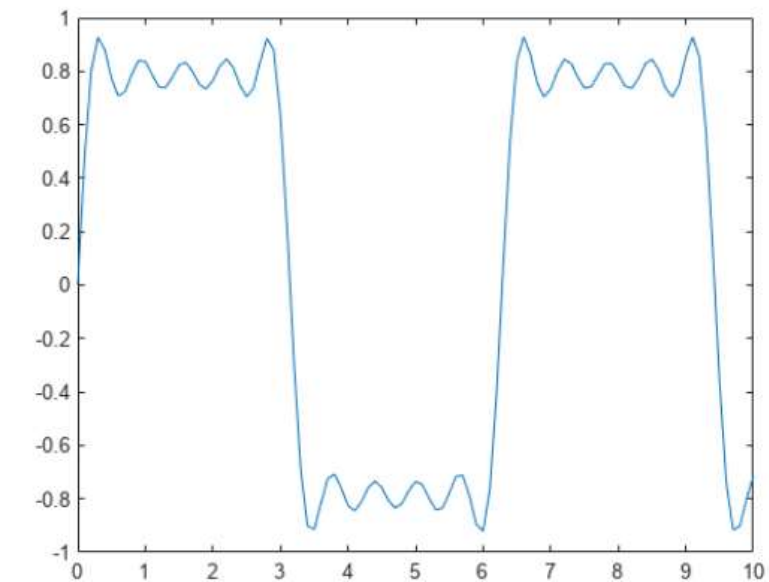
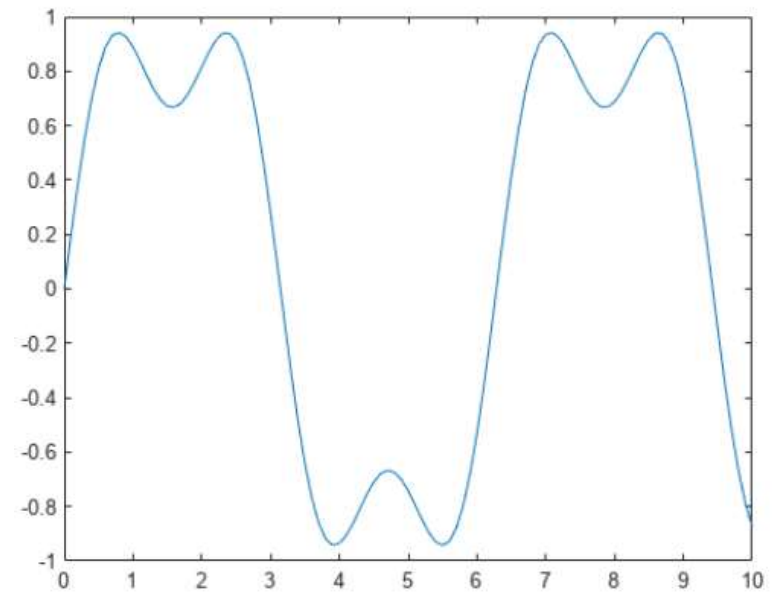
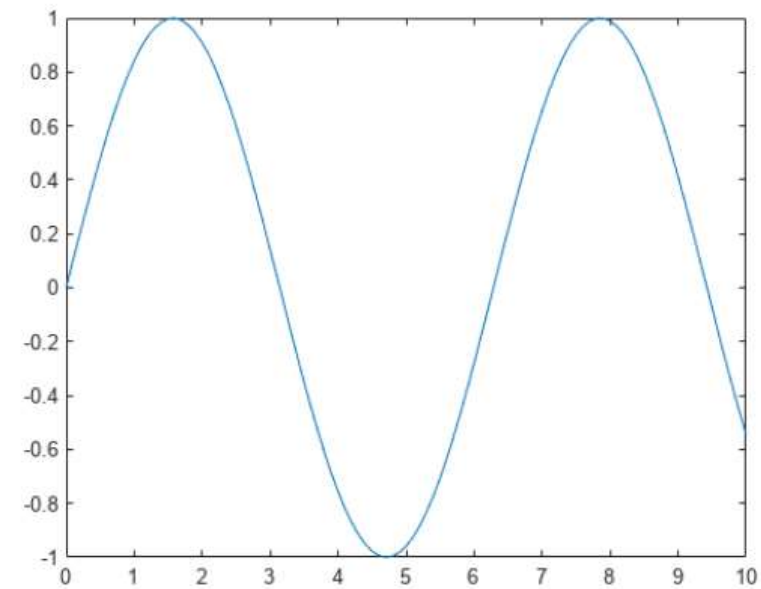
%y=fft2(p);
figure,surf(p);
%figure,surf(fftshift(real(y)));

```

Fourier transform basis functions

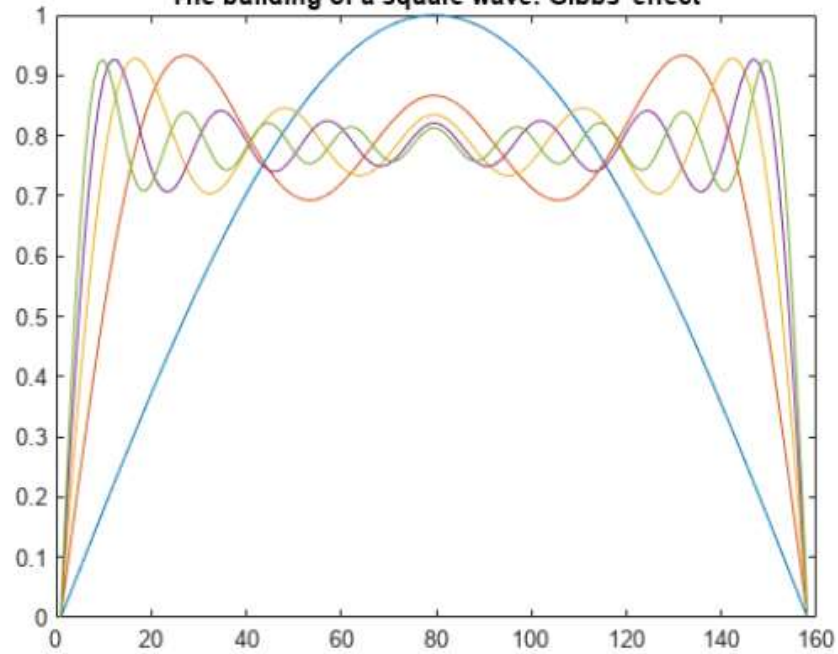
Approximating a
square wave as the
sum of sine waves.



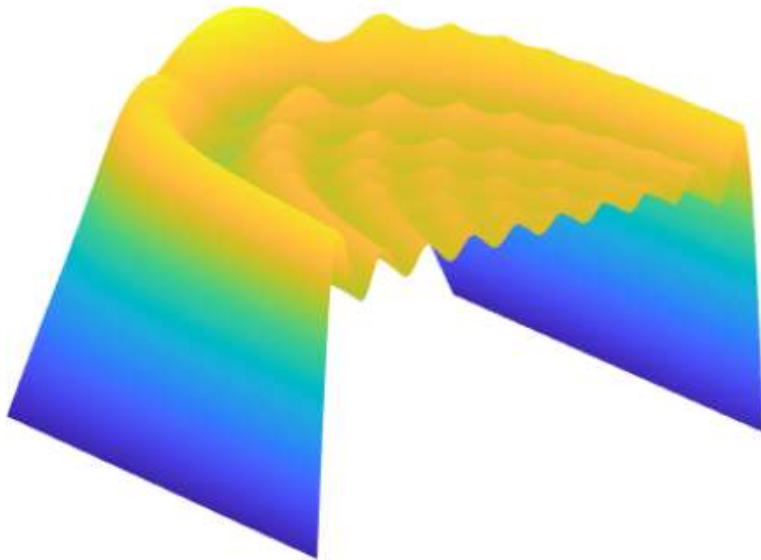


```
t = 0:.1:10;
y = sin(t);
plot(t,y);
y = sin(t) + sin(3*t)/3;
figure,plot(t,y);
y = sin(t) + sin(3*t)/3 + sin(5*t)/5 + sin(7*t)/7 + sin(9*t)/9;
figure,plot(t,y);
```

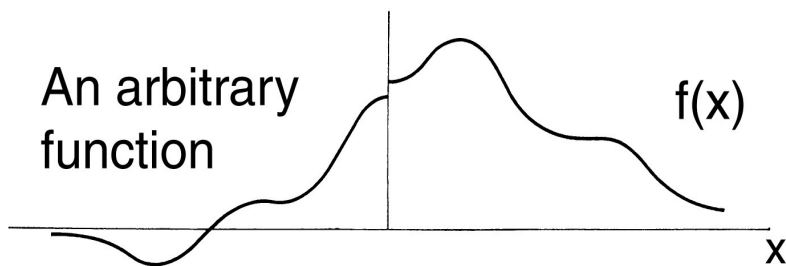
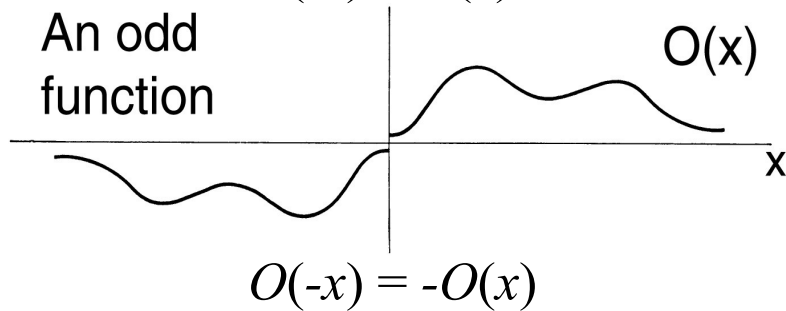
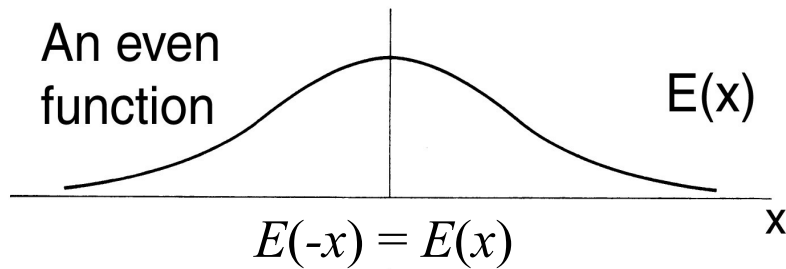

The building of a square wave: Gibbs' effect



```
t = 0:.02:3.14;  
y = zeros(10,length(t));  
x = zeros(size(t));  
for k = 1:2:19  
    x = x + sin(k*t)/k;  
    y((k+1)/2,:) = x;  
end  
plot(y(1:2:9,:))  
title('The building of a square wave: Gibbs'' effect');  
surf(y);  
shading interp  
axis off ij
```



Any function can be written as the
sum of an even and an odd function

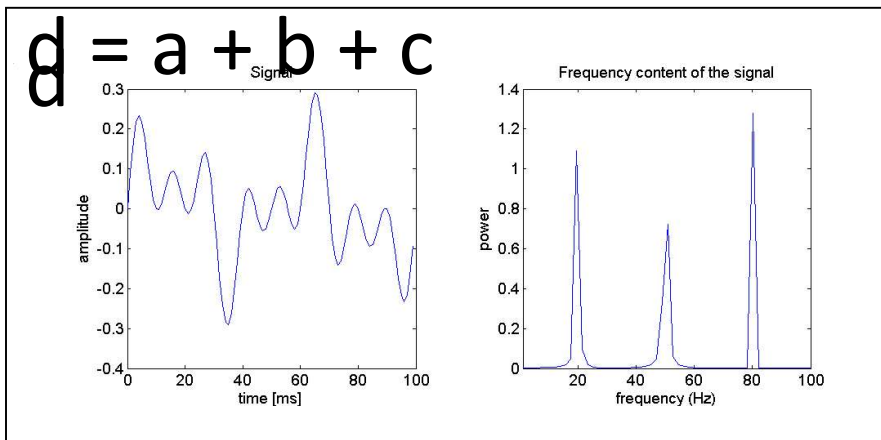


$$E(x) \equiv [f(x) + f(-x)] / 2$$

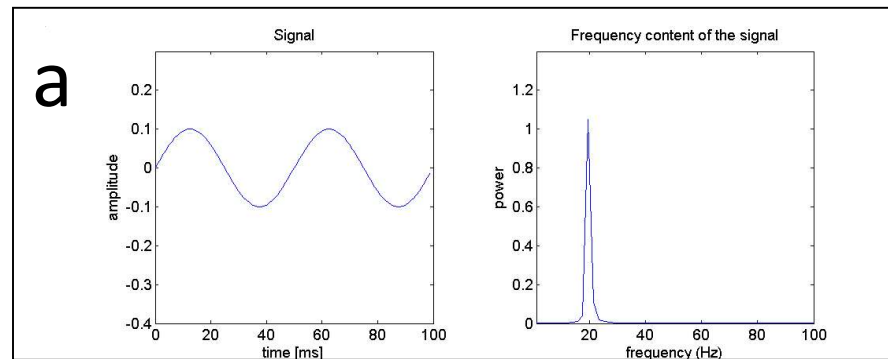
$$O(x) \equiv [f(x) - f(-x)] / 2$$

$\Downarrow$

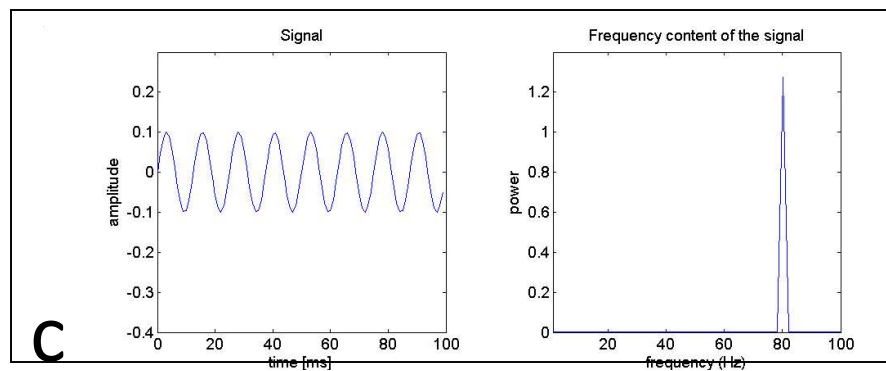
$$f(x) = E(x) + O(x)$$



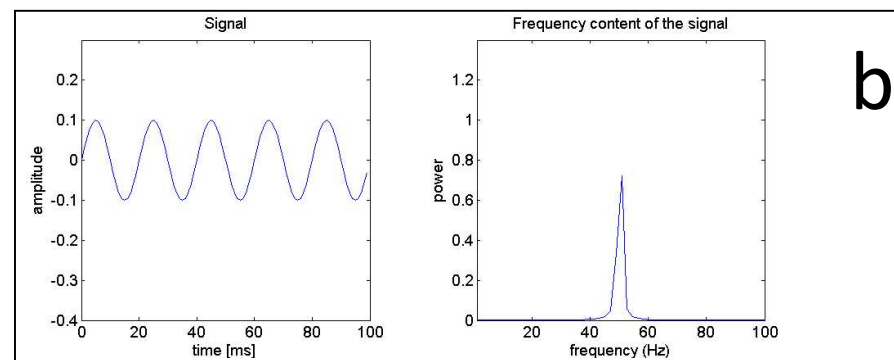
time domain frequency domain



time domain frequency domain



time domain frequency domain



time domain frequency domain

Complex Number

A complex number, C , is defined as

$$C = R + jI \quad (4.2-1)$$

where R and I are real numbers, and j is an imaginary number equal to the square of -1 ; that is, $j = \sqrt{-1}$. Here, R denotes the *real part* of the complex number and I its *imaginary part*. Real numbers are a subset of complex numbers in which $I = 0$. The *conjugate* of a complex number C , denoted C^* , is defined as

$$C^* = R - jI \quad (4.2-2)$$

Complex Number

Complex numbers can be viewed geometrically as points in a plane (called the *complex plane*) whose abscissa is the *real axis* (values of R) and whose ordinate is the *imaginary axis* (values of I). That is, the complex number $R + jI$ is point (R, I) in the rectangular coordinate system of the complex plane.

Sometimes, it is useful to represent complex numbers in polar coordinates,

$$C = |C|(\cos \theta + j \sin \theta) \quad (4.2-3)$$

where $|C| = \sqrt{R^2 + I^2}$ is the length of the vector extending from the origin of the complex plane to point (R, I) , and θ is the angle between the vector and the real axis. Drawing a simple diagram of the real and complex axes with the vector in the first quadrant will reveal that $\tan \theta = (I/R)$ or $\theta = \arctan(I/R)$. The $\arctan$ function returns angles in the range $[-\pi/2, \pi/2]$. However, because I and R can be positive and negative independently, we need to be able to obtain angles in the full range $[-\pi, \pi]$. This is accomplished simply by keeping track of the sign of I and R when computing θ . Many programming languages do this automatically via so called *four-quadrant arctangent* functions. For example, MATLAB provides the function `atan2(Imag, Real)` for this purpose.

Using Euler's formula,

$$e^{j\theta} = \cos \theta + j \sin \theta \quad (4.2-4)$$

Complex Number

$$C = |C|e^{j\theta}$$

$F(u) = R(u) + jI(u)$, where $R(u)$ and $I(u)$ are the real and imaginary component functions. As previously noted, the complex conjugate is $F^*(u)$

Complex Exponential(Euler Formula)

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$i = \sqrt{-1}$$

Expand $e^{i\theta}$ using **Taylor Series**:

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \frac{(i\theta)^6}{6!} + \dots$$

$$e^{i\theta} = \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right)}_{\cos \theta} + i \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots\right)}_{\sin \theta}$$

The Inverse Fourier Transform

We go from $f(t)$ to $F(u)$ by

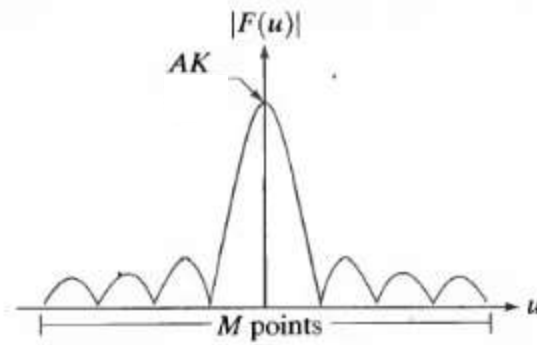
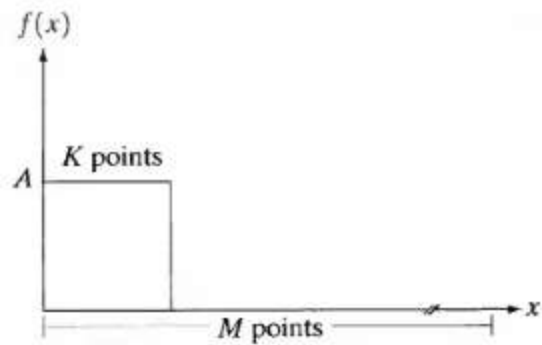
$$\mathfrak{F}\{f(t)\} = F(u) = \int_{-\infty}^{\infty} f(t) \exp(-j2\pi ut) dt$$

Fourier Transform

Given $F(u)$, $f(t)$ can be obtained by the inverse Fourier transform

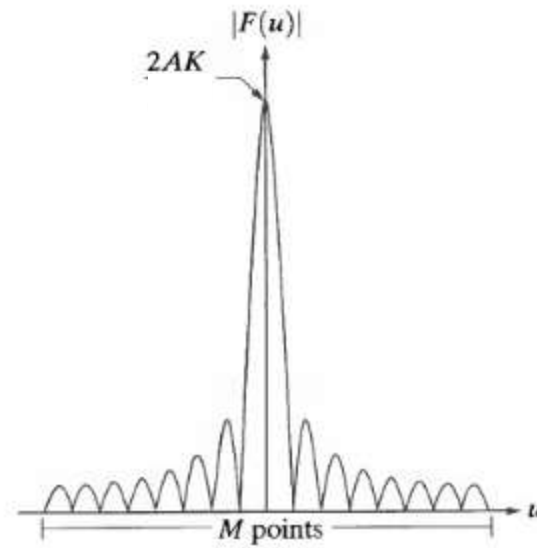
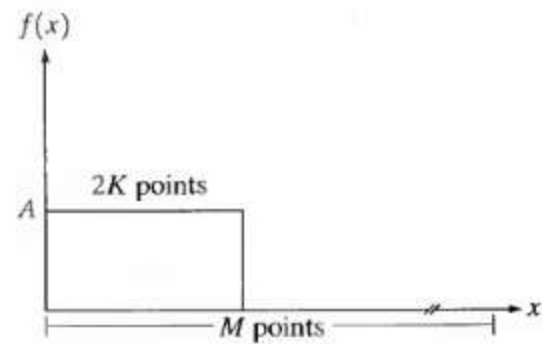
$$\mathfrak{F}^{-1}\{F(u)\} = f(t) = \int_{-\infty}^{\infty} F(u) \exp(j2\pi ut) du$$

Inverse Fourier Transform



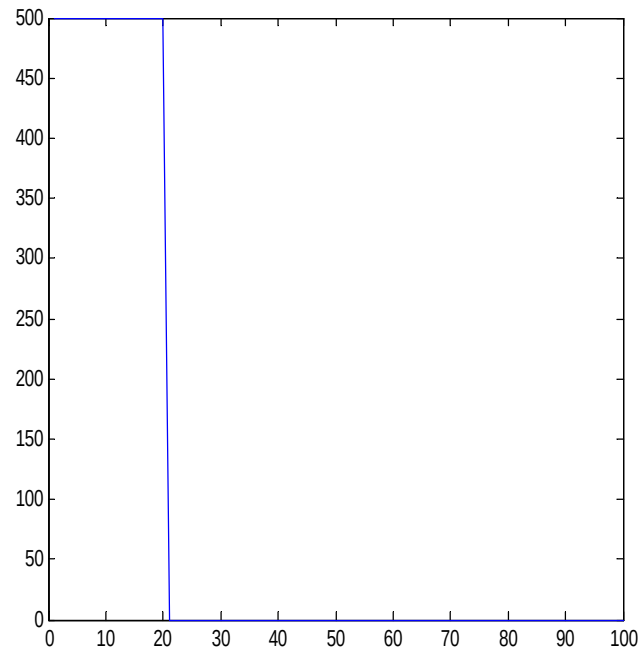
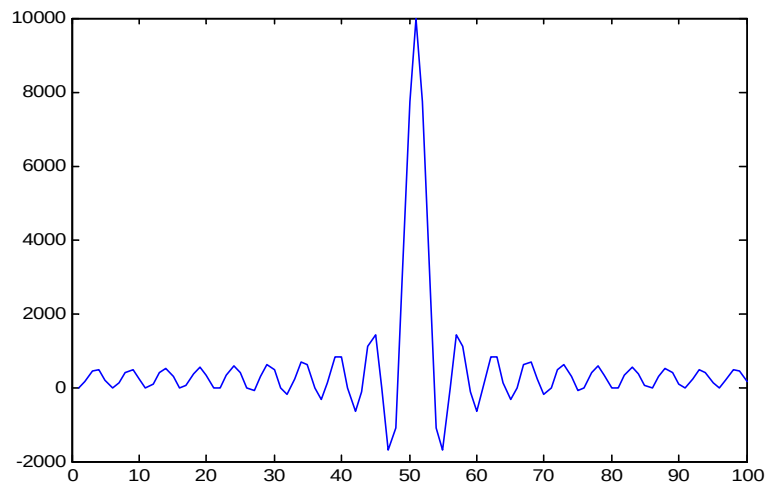
a b
c d

FIGURE 4.2 (a) A discrete function of M points, and (b) its Fourier spectrum. (c) A discrete function with twice the number of nonzero points, and (d) its Fourier spectrum.



$$\Delta u \cdot \Delta x \leq 1/M$$

```
a=zeros(1,100);  
a(0:20)=500;  
figure, plot(a);  
b=fft(a);  
c=abs(fftshift(b));  
figure,plot(c);
```



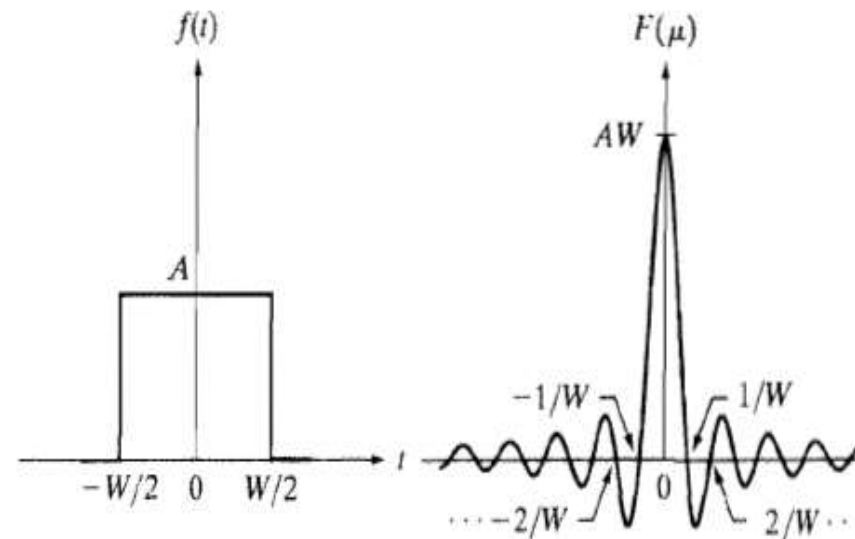
Use the identities

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

Fourier Transform

$$\begin{aligned}
 F(\mu) &= \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt = \int_{-W/2}^{W/2} A e^{-j2\pi\mu t} dt \\
 &= \frac{-A}{j2\pi\mu} \left[e^{-j2\pi\mu t} \right]_{-W/2}^{W/2} = \frac{-A}{j2\pi\mu} \left[e^{-j\pi\mu W} - e^{j\pi\mu W} \right] \\
 &= \frac{A}{j2\pi\mu} \left[e^{j\pi\mu W} - e^{-j\pi\mu W} \right] \\
 &= AW \frac{\sin(\pi\mu W)}{(\pi\mu W)}
 \end{aligned}$$



a b c

FIGURE 4.4 (a) A simple function; (b) its Fourier transform;

- The result in the last step of the preceding expression is known as *sinc* function:

$$\text{sinc}(m) = \frac{\sin(\pi m)}{(\pi m)}$$

- Where $\text{sinc}(0) = 1$ and $\text{sinc}(m) = 0$ for all other integer values of m .

Fourier Transform of Cosine Function

Given

$$x(t) = \cos \omega_0 t$$

From Euler's rule, we have,

$$\cos \omega_0 t = \left[\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right]$$

Then, from the definition of Fourier transform, we have,

$$F[\cos \omega_0 t] = X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = \int_{-\infty}^{\infty} \cos \omega_0 t e^{-j\omega t} dt$$

$$\Rightarrow X(\omega) = \int_{-\infty}^{\infty} \left[\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right] e^{-j\omega t} dt$$

$$\begin{aligned} \Rightarrow X(\omega) &= \frac{1}{2} \left[\int_{-\infty}^{\infty} e^{j\omega_0 t} e^{-j\omega t} dt + \int_{-\infty}^{\infty} e^{-j\omega_0 t} e^{-j\omega t} dt \right] \\ &= \frac{1}{2} \{F[e^{j\omega_0 t}] + F[e^{-j\omega_0 t}]\} \end{aligned}$$

Fourier Transform of cosine function

$$\Rightarrow X(\omega) = \frac{1}{2}[2\pi\delta(\omega - \omega_0) + 2\pi\delta(\omega + \omega_0)]$$

$$\Rightarrow X(\omega) = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

Therefore, the Fourier transform of cosine wave function is,

$$F[\cos \omega_0 t] = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

Or, it can also be represented as,

$$\cos \omega_0 t \xleftrightarrow{\text{FT}} \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

The graphical representation of the cosine wave signal with its magnitude and phase spectra is shown in Figure-2.

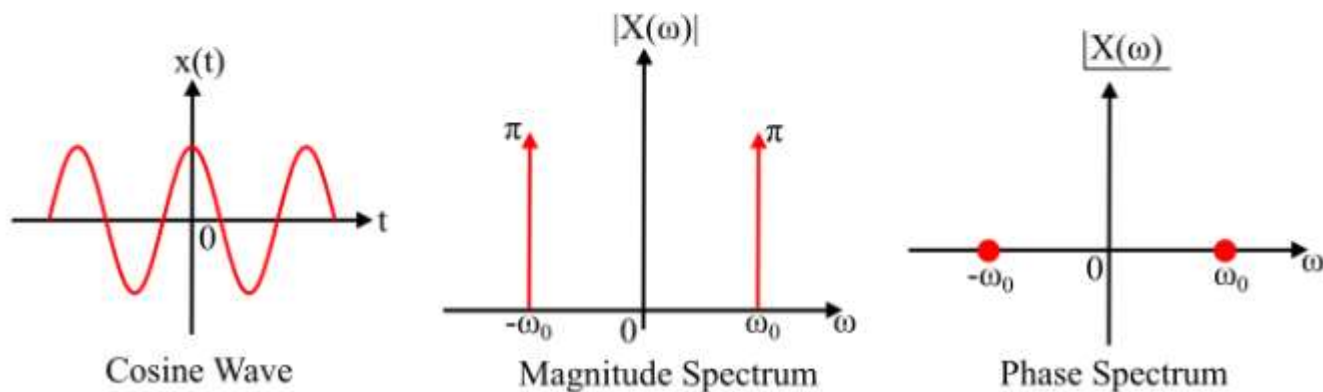


Figure-2

Unit impulse of a continuous variable t located at $t=0$:

$$\delta(t) = \begin{cases} \infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0 \end{cases}$$

and

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

Sifting property :

$$\int_{-\infty}^{\infty} f(t) \delta(t) dt = f(0)$$

(if $f(t)$ continuous at $t=0$)

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0)$$

Unit discrete impulse located at $x=0$:

$$\delta(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases}$$

$$\sum_{x=-\infty}^{\infty} \delta(x) = 1$$

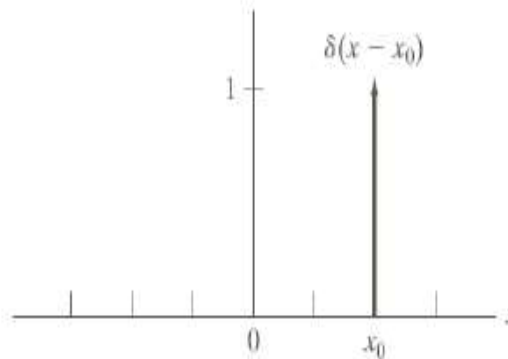


FIGURE 4.2

A unit discrete impulse located at $x = x_0$. Variable x is discrete, and δ is 0 everywhere except at $x = x_0$.

Sifting property :

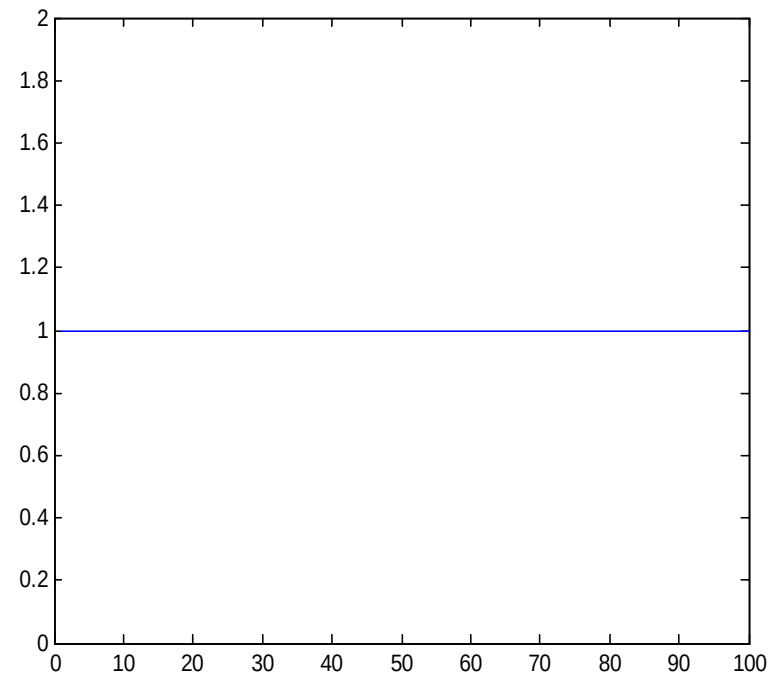
$$\sum_{x=-\infty}^{\infty} f(x)\delta(x) = f(0)$$

$$\sum_{x=-\infty}^{\infty} f(x)\delta(x - x_0) = f(x_0)$$

Fourier Transform of Unit Impulse located at origin

$$\begin{aligned} F(\mu) &= \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi\mu t} dt \\ &= \int_{-\infty}^{\infty} e^{-j2\pi\mu t} \delta(t) dt \\ &= e^{-j2\pi\mu 0} = e^0 \\ &= 1 \end{aligned}$$

```
a=zeros(1,100);  
a(1,1)=1;  
figure, plot(a);  
b=fft(a);  
c=abs(fftshift(b));  
figure,plot(c);
```



Fourier Transform of Impulse located at $t=t_0$

$$\begin{aligned} F(\mu) &= \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j2\pi\mu t} dt \\ &= \int_{-\infty}^{\infty} e^{-j2\pi\mu t} \delta(t - t_0) dt \\ &= e^{-j2\pi\mu t_0} \\ &= \cos(2\pi\mu t_0) - j \sin(2\pi\mu t_0) \end{aligned}$$